

Jaeseung Lee

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Software engineer with experience building reproducible data processing workflows, analytical pipelines, and research software for large-scale biomedical datasets. Skilled in Python, R, Bash, Linux, Git, data quality control, and structured documentation. Experienced in translating research needs into technical solutions and communicating complex technical findings to physician-scientists, researchers, and non-technical stakeholders.

EXPERIENCE

DIGITAL PATHOLOGY WEB PORTAL

2026 (Ongoing)

Independent Project | [\[Demo\]](#) [\[GitHub\]](#)

- Built a web-based research data product for reviewing large biomedical imaging datasets, including structured case navigation, slide selection, metadata display, and interactive image exploration.
- Translated pathology research workflow needs into frontend features that support case-level review, dataset navigation, and efficient data exploration.

COLUMBIA UNIVERSITY IRVING MEDICAL CENTER

New York, NY

Department of Pathology and Cell Biology

Staff Associate I

Nov 2023 – Jun 2024

- Processed and analyzed NGS-based transcriptomic datasets, including single-cell RNA-seq and spatial transcriptomics; performed quality control, batch correction, cell-type annotation, differential gene expression analysis, and pathway enrichment analysis.
- Developed computational pipelines for spatial transcriptomics analysis, including cell-cell interaction modeling and transcriptomic characterization, achieving up to 1200× speedup through GPU acceleration; co-authored a *Nature Communications* publication and released the code on GitHub. [\[Paper\]](#) [\[GitHub\]](#)

Visiting Researcher

Apr 2023 – Jun 2023

- Managed project-level data tracking for raw sequencing files, processed transcriptomics datasets, and image files; maintained structured spreadsheets documenting file locations, sample metadata, and processing status.

KOREA UNIVERSITY OF TECHNOLOGY AND EDUCATION

South Korea

Department of Computer Engineering

Research Assistant

Mar 2021 – Dec 2022

- Developed an AI system to detect physiological responses to music using wearable sensor data, achieving 81% prediction accuracy; co-authored a peer-reviewed publication.

SKILL

- **Programming:** Python, R, C/C++, Bash
- **Operating System:** Linux (Ubuntu)
- **Version Control:** Git/GitHub
- **Cloud:** Amazon Web Service (AWS)
- **AI Code Agent:** Cursor, Codex
- **AI/ML:** TensorFlow, PyTorch, Scikit-learn

PUBLICATION

- Ma, M., Paryani, F., Jakubiak, K., Xia, S., Antoku, S., Kannan, A., **Lee, J.**, ... Sims, P. A. & Al-Dalahmah, O. *The spatial landscape of glial pathology and T cell response in Parkinson's disease substantia nigra. Nature Communications*. 2025; 16:7146. [\[Paper\]](#) [\[GitHub\]](#)
- Lee, E., Min, C., **Lee, J.**, Yu, J. and Kang, S. *Groovemeter: Enabling music engagement-aware apps by detecting reactions to daily music listening via Earable sensing. Proc. 31st ACM Int. Conf. on Multimedia*. 2023. [\[Paper\]](#)

EDUCATION

KOREA UNIVERSITY OF TECHNOLOGY AND EDUCATION

South Korea

Bachelor of Science in Electrical Engineering

Feb 2021